

LIMITS

- **Limit:** Let $f(x)$ be defined on an interval about 'a' except at 'a' itself. We say $f(x)$ approaches to a limit L as x approaches to 'a' and written as $\lim_{x \rightarrow a} f(x) = L$, if for every number $\epsilon > 0$, there exist a corresponding number $\delta > 0$ such that for all x , $|x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$.
- **How to find a δ for a given $f(x), L, x_0$, and $\epsilon > 0$ Algebraically:**
 - (1) Solve the inequality $|f(x) - L| < \epsilon$ to find an open interval (a, b) about x_0 on which the inequality holds for all $x \neq x_0$.
 - (2) Find the value of $\delta > 0$ that places the open interval $(x_0 - \delta, x_0 + \delta)$ centred at x_0 inside the interval (a, b) . The inequality $|f(x) - L| < \epsilon$ will hold for all $x \neq x_0$ in this interval $(x_0 - \delta, x_0 + \delta)$.
- **Left Hand Limit:** The left hand limit $\lim_{x \rightarrow a^-} f(x) = L$, if x approaches a from the left side of a , that is it crosses over values of the form $a - h$, $h > 0$, the difference between $f(x)$ and L can be made arbitrarily small. More precisely $\lim_{x \rightarrow a^-} f(x) = L$, if for every number $\epsilon > 0$, there exist a corresponding number $\delta > 0$ such that for $a - \delta < x < a$, then $|f(x) - L| < \epsilon$. Or it can be calculated as $\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a - h)$.
- **Right Hand Limit:** The right hand limit $\lim_{x \rightarrow a^+} f(x) = L$, if x approaches a from the right side of a , that is it crosses over values of the form $a + h$, $h > 0$, the difference between $f(x)$ and L can be made arbitrarily small. More precisely $\lim_{x \rightarrow a^+} f(x) = L$, if for every number $\epsilon > 0$, there exist a corresponding number $\delta > 0$ such that for $a < x < a + \delta$, then $|f(x) - L| < \epsilon$. Or it can be calculated as $\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a + h)$.
- **Neighborhood of a point a :** Let a be any real number. Then for a positive real number $\delta > 0$, the open interval $(a - \delta, a + \delta)$ is called as neighborhood of a .
- **Left Neighborhood of a point a :** Let a be any real number. Then for a positive real number $\delta > 0$, the open interval $(a - \delta, a)$ is called as left neighborhood of a .
- **Right Neighborhood of a point a :** Let a be any real number. Then for a positive real number $\delta > 0$, the open interval $(a, a + \delta)$ is called as right neighborhood of a .
- **Deleted Neighborhood of a point a :** Let a be any real number. Then for a positive real number $\delta > 0$, the open interval $(a - \delta, a + \delta) - \{a\}$ is called as deleted neighborhood of a .
- **Indeterminate Forms:** If any function f is in any of the forms $\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 1^\infty, 0^0, \infty^0$ at $x = a$, then function f is in indeterminate form at $x = a$.
- **Formula related to Limits:**

- (a) $\lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x).$
- (b) $\lim_{x \rightarrow a} (f(x) \times g(x)) = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x).$
- (c) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \quad (g(x) \neq 0).$
- (d) $\lim_{x \rightarrow a} (cf(x)) = c \lim_{x \rightarrow a} f(x),$ where c is a constant.
- (e) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1.$
- (f) $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1.$
- (g) $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = \lim_{x \rightarrow 0} \frac{x}{\sin^{-1} x} = 1.$
- (h) $\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = \lim_{x \rightarrow 0} \frac{x}{\tan^{-1} x} = 1.$
- (i) $\lim_{x \rightarrow 0} \cos x = 1.$
- (j) $\lim_{x \rightarrow a} \sin^{-1} x = \sin^{-1} a,$ where $|a| \leq 1.$
- (k) $\lim_{x \rightarrow a} \cos^{-1} x = \cos^{-1} a,$ where $|a| \leq 1.$
- (l) $\lim_{x \rightarrow a} \tan^{-1} x = \tan^{-1} a.$
- (m) $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{x} = 0.$
- (n) $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}.$
- (o) $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1.$
- (p) $\lim_{x \rightarrow e} \log_e x = 1.$
- (q) $\lim_{x \rightarrow 0} \frac{\log(1-x)}{x} = -1.$
- (r) $\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \log_a e.$
- (s) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$
- (t) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_a e.$
- (u) $\lim_{x \rightarrow 0} (1+x)^{1/x} = e.$
- (v) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$

$$(w) \lim_{n \rightarrow \infty} x^n = \begin{cases} \infty & x > 1 \\ 1 & x = 1 \\ 0 & -1 < x < 1 \\ \text{does not exist} & x \leq -1 \end{cases}$$

$$(x) \lim_{n \rightarrow \infty} x^{2n} = \begin{cases} \infty & x > 1 \\ 1 & x = \pm 1 \\ 0 & -1 < x < 1 \\ \infty & x < -1 \end{cases}$$

$$(y) \lim_{n \rightarrow \infty} x^{2n+1} = \begin{cases} \infty & x > 1 \\ 1 & x = 1 \\ 0 & -1 < x < 1 \\ -1 & x = -1 \\ -\infty & x < -1 \end{cases}$$

$$(z) \text{ If } \lim_{x \rightarrow a} f(x) = 1, \lim_{x \rightarrow a} g(x) = \infty, \text{ then } \lim_{x \rightarrow a} (f(x))^{g(x)} = \lim_{x \rightarrow a} (1 + f(x) - 1)^{g(x)} = e^{\lim_{x \rightarrow a} (f(x) - 1)g(x)}.$$

$$(aa) \text{ If } \lim_{x \rightarrow a} (f(x))^{g(x)} \text{ is } 0^0 \text{ or } \infty^0 \text{ then } \lim_{x \rightarrow a} (f(x))^{g(x)} = e^{\lim_{x \rightarrow a} g(x) \log f(x)}.$$

- **L' Hospital Rule:** Let $f(x)$ and $g(x)$ are two functions differentiable in some neighborhood of the point a , except may be at a itself. If $\lim_{x \rightarrow a} f(x) = 0, \lim_{x \rightarrow a} g(x) = 0$ or

$$\lim_{x \rightarrow a} f(x) = \infty, \lim_{x \rightarrow a} g(x) = \infty, \text{ then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

- **Newton-Leibnitz's Formula in Evaluating the Limits:** Let us consider the integral if

$$l(x) = \int_a^{g(x)} f(t) dt \Rightarrow \frac{dl(x)}{dx} = f(g(x)) \cdot \frac{dg(x)}{dx}$$

$$l(x) = \int_{h(x)}^{g(x)} f(t) dt \Rightarrow \frac{dl(x)}{dx} = f(g(x)) \cdot \frac{dg(x)}{dx} - f(h(x)) \cdot \frac{dh(x)}{dx}.$$

- **Some Important Series:**

$$(a) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$(b) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$(c) \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$$

- (d) $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$
- (e) $\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$
- (f) $\tanh x = x - \frac{x^3}{3} + \frac{2x^5}{15} - \frac{17x^7}{315} + \dots$
- (g) $\sin^{-1} x = x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \dots$
- (h) $\cos^{-1} x = \frac{\pi}{2} - \left(x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \dots \right)$
- (i) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$
- (j) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$
- (k) $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \dots$
- (l) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$
- (m) $a^x = e^{x \log a} = 1 + \frac{(x \log a)}{1!} + \frac{(x \log a)^2}{2!} + \frac{(x \log a)^3}{3!} + \frac{(x \log a)^4}{4!} + \frac{(x \log a)^5}{5!} + \dots$
